#3 反函数求导原理揭秘

设$y=f(x)$单调可导且$f'(x)\neq 0$，则$y=f(x)$的反函数$x=\varphi(y)$也可导，则

$$\frac{\mathrm{d}x}{\mathrm{d}y}=\frac{1}{\mathrm{d}y/\mathrm{d}x}=\frac{1}{f'(x)},\quad \frac{\mathrm{d}^2x}{\mathrm{d}y^2}=\frac{\mathrm{d}}{\mathrm{d}y}\left(\frac{\mathrm{d}x}{\mathrm{d}y}\right)=\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{f'(x)}\right)\frac{\mathrm{d}x}{\mathrm{d}y}=-\frac{f''(x)}{\left[f'(x)\right]^3}.$$

设$f(x)$严格单调可导，$f'(x)\neq 0$，且$f(x)$在$x=1$处二阶可导，$f(1)=-2$，$f'(1)=-\dfrac{\sqrt{2}}{2}$，$f''(1)=2$，记$y=f(x)$的反函数为$x=\varphi(y)$，求$\left.\dfrac{\mathrm{d}x}{\mathrm{d}y}\right|_{y=-2}$，$\left.\dfrac{\mathrm{d}^2x}{\mathrm{d}y^2}\right|_{y=-2}$.

【答案】

因为$f(1)=-2$，当$y=-2$时$x=1$，已知$f'(1)=-\frac{\sqrt{2}}{2}$，

所以$\frac{dx}{dy}|_{y=-2}=\frac{1}{f'(1)}=-\sqrt{2}$.

由复合函数求导法则$\frac{d}{dy}\left(\frac{1}{f'(x)}\right)=\frac{d}{dx}\left(\frac{1}{f'(x)}\right)\cdot\frac{dx}{dy}$.

$\frac{d}{dx}\left(\frac{1}{f'(x)}\right)=-\frac{f''(x)}{(f'(x))^2}$，又$\frac{dx}{dy}=\frac{1}{f'(x)}$，所以$\frac{d^2x}{dy^2}=-\frac{f''(x)}{(f'(x))^3}$.

将$x=1, f'(1)=-\frac{\sqrt{2}}{2}, f''(1)=2$代入得：

$$\frac{d^2x}{dy^2}|_{y=-2}=-\frac{2}{\left(-\frac{\sqrt{2}}{2}\right)^3}=4\sqrt{2}.$$